Discovery Log – Calculator App Setup

Project: Expo React Native Calculator App
Environment: Windows 11, Cursor IDE, Android Studio Emulator (Pixel 5 API 33)
Date: April 1–2, 2026
Status: ✅ Completed Successfully

Overview

This log documents all procedures, errors encountered, and discoveries made while setting up and running a React Native calculator app using Expo and Cursor AI IDE on Windows. The purpose is to serve as a reference guide for future mobile app project setups in the same environment.

Phase 1: Environment Setup

Node.js and npx

`npx` does not need to be installed separately. It ships bundled with `npm`, which ships with **Node.js**. The correct procedure to get `npx` working is to install Node.js LTS from nodejs.org, after which all three tools — `node`, `npm`, and `npx` — become available from the terminal.[1]

Discovery: On Windows, if `npx` throws a `.ps1 cannot be loaded because running scripts is disabled` error in PowerShell, the fix is to run once in an Administrator PowerShell:[2]

```
Set-ExecutionPolicy -Scope CurrentUser -ExecutionPolicy RemoteSigned
```

Alternatively, using **Command Prompt (cmd.exe)** instead of PowerShell bypasses the execution policy restriction entirely.[2]

Android Studio and Emulator

The Android emulator used throughout this project was a **Pixel 5, API 33** virtual device created inside Android Studio. The emulator must be running before `npx expo run:android` is issued, but it does not need to be manually started again for subsequent runs — Expo will target the open emulator automatically.[3]

Key Android paths on this machine:

- **Android SDK:** `C:\Users\ethan\AppData\Local\Android\Sdk`
- **ADB binary:** `C:\Users\ethan\AppData\Local\Android\Sdk\platform-tools\adb.exe`

Phase 2: Creating the Expo Project

Creating the App in Cursor

Cursor AI was used to scaffold the project. Cursor ran `npx create-expo-app@latest` internally to create the app structure. The following structure was generated:[4]

```
Calculator_App/
└── calculator_app/
    ├── App.tsx
    ├── app.json
    ├── index.ts
    ├── tsconfig.json
    ├── package.json
    ├── assets/
    ├── src/
    │   └── calculatorLogic.ts
    ├── android/
    └── node_modules/
```

Bug 1 — Running `npm install` in the wrong folder

Error:

```
npm error enoent Could not read package.json
```

Cause: `npm install` was run inside the parent `Calculator_App` folder, which had no `package.json`. The Expo project and its `package.json` live inside the inner `calculator_app` subfolder.[5][6]

Fix: Always `cd` into `calculator_app` (the inner folder) before running any npm or npx commands.

Bug 2 — Folder name with spaces

Error:

```
Cannot create an app named "Calculator App". The project name can only co
```

Cause: `create-expo-app` derives the project name from the folder name. A folder named `Calculator App` with a space is not URL-friendly.[7][8]

Fix: Rename the folder to use hyphens or underscores — `calculator-app` or `calculator_app` — before running `npm init expo@latest .`.[9][8]

Bug 3 — Trying to re-scaffold an existing project

Error:

```
The directory calculator-app has files that might be overwritten: app.json, Ap
Try using a new directory name, or moving these files.
```

Cause: Cursor had already scaffolded the project; running `npm init expo@latest .` again inside the same folder tried to overwrite existing files.[10]

Fix: Do not re-run `npm init expo@latest .` if the folder already contains a valid Expo project (`package.json`, `App.tsx`, `node_modules`).

Verify with `dir` and proceed directly to `npm install` then `npx expo run:android`.[10]

Phase 3: Building and Running

Bug 4 — Android SDK not found (Gradle build failure)

Error:

```
SDK location not found. Define a valid SDK location with an ANDROID_HOM
or by setting the sdk.dir path in your project's local properties file at
'...calculator_app\android\local.properties'.
```

Cause: The Expo prebuild generated the `android/` folder, but Gradle could not locate the Android SDK because neither `ANDROID_HOME` was set as a system environment variable, nor did `android/local.properties` exist.[11][12]

Fix: Create the file `calculator_app/android/local.properties` manually (it is not auto-generated) and add:

```
sdk.dir=C\:\\Users\\ethan\\AppData\\Local\\Android\\Sdk
```

Note: On Windows, backslashes in `local.properties` must be escaped with a second backslash.[12][11]

Alternative permanent fix: Set `ANDROID_HOME` as a Windows system environment variable pointing to `C:\Users\ethan\AppData\Local\Android\Sdk`. This eliminates the need for `local.properties` on future projects.[13]

Bug 5 — Legacy global expo-cli warning

Error/Warning:

```
The global expo-cli package has been deprecated.
Cannot determine which native SDK version your project uses because the r
```

Cause: A globally installed legacy `expo-cli` was being invoked instead of the local Expo CLI bundled inside `node_modules`.[14]

Fix: Always use `npx expo mmand>` (e.g., `npx expo run:android`) from inside the project folder. This invokes the local Expo CLI in `node_modules`, not the deprecated global one.[14]

Bug 6 — TypeScript compiler not found

Error:

```
This is not the tsc command you are looking for.
Use npm install typescript to first add TypeScript to your project before usin
```

Cause: TypeScript was not yet installed as a dev dependency when `npx tsc --noEmit` was run.[15]

Fix:

```
npm install
npm install -D typescript
npx tsc --noEmit
```

Note: Running `npx tsc --noEmit` to type-check is optional and not required to run the app. `npx expo run:android` can be run directly.[16]

Phase 4: Emulator Behavior Discoveries

Discovery 1 — Expo launches the app automatically

Finding: `npx expo run:android` handles the full pipeline: build → install → launch. There is no need to manually open the app on the emulator. The command targets any open emulator automatically.[14]

Procedure: Open the Android Studio emulator first, then run `npx expo run:android` from the project folder and wait for it to finish. Do not interrupt the process.

Discovery 2 — Black emulator screen is not always a failure

Finding: After `npx expo run:android` reported a successful bundle and install, the emulator appeared black. The app had actually launched correctly, but the emulator display was in a sleep/locked state.[17][18]

Fix: Press the **power button** on the emulator toolbar to wake the screen. The app appeared immediately after waking.[17]

Log note: A black emulator screen does not mean the build failed. Always press the emulator power button before assuming failure.

Discovery 3 — White screen is a runtime/app issue, not an environment issue

Finding: A recurring white screen on the emulator after the app installed indicated a JavaScript startup error or render failure, not an environment or build problem. This was repeatedly confirmed to be separate from the installation/bundling process.[19][20][21]

Troubleshooting checklist for white screen:

1. Run `adb logcat -s ReactNativeJS` while reloading the app to capture JS errors.
2. Run `npx expo start --clear` to clear Metro bundler cache.
3. Confirm all packages are installed with `npm install`.
4. Rebuild with `npx expo run:android` after any package changes.

Phase 5: Final Working Procedure

Once the environment was fully configured, the reliable procedure to run the calculator app is:

1. Open Android Studio and start the **Pixel 5 API 33** emulator.
2. Navigate to the project in PowerShell or Command Prompt:

   ```
   cd "C:\Users\ethan\Personal Projects\Mobile App Projects\Calculator_Ap
   ```

3. Install dependencies (first time only):

```
npm install
```

4. Build and run:

```
npx expo run:android
```

5. Wait for the build to complete. If the emulator screen is black, press the power button to wake it.

Summary of Key Rules

Rule	Details
Always work inside `calculator_app`	Not the parent `Calculator_App` folder
Folder names must be URL-friendly	No spaces; use hyphens or underscores
`npx` comes with Node.js	Install Node.js to get `npm` and `npx`
Create android/local.properties manually	Set sdk.dir to your Android SDK path
Use `npx expo` not global `expo-cli`	Legacy global CLI is deprecated
Black emulator screen = press power button	App may have launched but screen is asleep
White screen = check JS errors via `adb logcat`	Not an environment issue
Never re-scaffold an existing project	Check for package.json first with dir

---

References

1. Downloading and installing Node.js and npm - If you're using macOS or Windows, use one of the installers from the Node.js download page. Be sure ...
2. File C:\Program Files\nodejs\npx.ps1 cannot be loaded because ... - This error occurs because PowerShell has restricted script execution for security reasons. Steps to ...
3. Android Studio Emulator - Expo Documentation - Open Android Studio, go to Settings > Languages & Frameworks > Android SDK. ... Copy adb from Androi...
4. create-expo-app - create-expo-app is a command-line tool to create and set up a new Expo and React Native project. Thi...
5. npm install module in current directory - node.js - Stack Overflow - The rule of thumb is: npm will not install a package's files directly into the current directory. Ho...
6. no such file or directory, open 'C:\Users\Nuwanst\package.json ... - Run this command first again. C:\Users\Nuwanst\Documents\NodeJS\3.chat>npm init It creates a package...
7. Create a project - Expo Documentation - Learn how to create a new Expo project ... We recommend starting with the default project created by...
8. Fix 'Sorry, name can only contain URL-friendly characters.' in npm - It requests “URL-friendly characters”, which means you need to name your package using URL-type form...
9. Characters to Avoid in Filenames and Directories - Don't include spaces or other punctuation in your folder or file name. · Try to keep your folder or ...
10. How do I solve the error called " the directory my-app contains files ... - The directory my-app contains files that could conflict: package.json. Either try using a new direct...
11. SDK location not found. Define location with sdk.dir in the local ... - Error: SDK location not found. Define location with sdk.dir in the local.properties file or with an ...
12. How to Fix "SDK location not found" in Android Studio? - Configure -> Project Defaults -> Project Structure -> SDKs on left column -> Android SDK Home Path -...

13. Environment variables | Android Studio - One of the most useful environment variables to set is ANDROID_HOME , which many tools read to deter...
14. Expo CLI - Expo Documentation - The npx expo start command automatically launches the app in a development build if expo-dev-client ...
15. Differences Between npm and npx - GeeksforGeeks - npx (Node Package Execute) is a tool that comes with npm (version 5.2. 0 and above) and is used to e...
16. Create your first app - Expo Documentation - In this tutorial, we'll build our app from scratch and understand the fundamentals of adding a file-...
17. Android emulator shows nothing except black screen and adb ... - Checking "Wipe user data" in the Launch Options fixed it for me. Go to Android Virtual Device Manage...
18. Troubleshoot known issues with Android Emulator | Android Studio - This document lists known issues, workarounds, and troubleshooting tips for common problems encounte...
19. Expo React native new EAS build blank white screen #29449 - GitHub - When i Created a APK build using EAS it just shows me white screen and there is also no logo of my a...
20. White screen on app load on production apps as of today #1378 - Seeing a white screen immediately after the splash screen on a standalone app built with expo build:...
21. react native application on emulator stuck on white screen - The only possibility coming to my mind is the network connectivity : make sure your emulator has act...